King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT222: Database Principles
1st Semester 1447 H

Hilton Honors

Section #	NAME	ID
View Name: Customer view		
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Project Description:

Hilton Honors is a hotel app with booking, rooms, payment, and points. It helps guests, staff, and managers to book, check in and out, and ask for services.
The goal of the database is to keep hotel data organized, make things easier for guests, and improve the service.

View Description:

The guest can use the app to search hotels, book or cancel, check in and out, open the room with digital key, ask for services, and see their points. The data includes bookings, room details, payments, and points.

Transaction Requirements:

Insert Transactions

1. The guest enters his ID, name,phone, and emails.
2. The guest enters the check-in and check-out dates.

Update/Delete Transactions
1. Update the phone number of a guest.
2. Update reservation dates (reschedule)
3. Delete a Reservation (cancels).

Retrieval Transactions

Without Condition

1. List all rooms.

2. List the number of my reservations.

3. List all rooms ordered by price from lowest to highest.
    a. With Condition
4. List all rooms of type ‘Suite’.

5. List rooms with prices less than 1000.

6. List my reservations for a specific room type and price range
    a. ex:This lists all reservations that are for “Deluxe” rooms and have a total price between 500 and 1000.
7. List my reservations between specific dates .

8. List all available rooms for a given date range.

9. List my reservations handled by a specific employee and within a total price range.
    a. ex:reservations processed by employee “Ali Ahmed” with total prices between 300 and 800
10. List my reservations with a count greater than 1.

11. List rooms with prices greater than average.

Data Requirements:

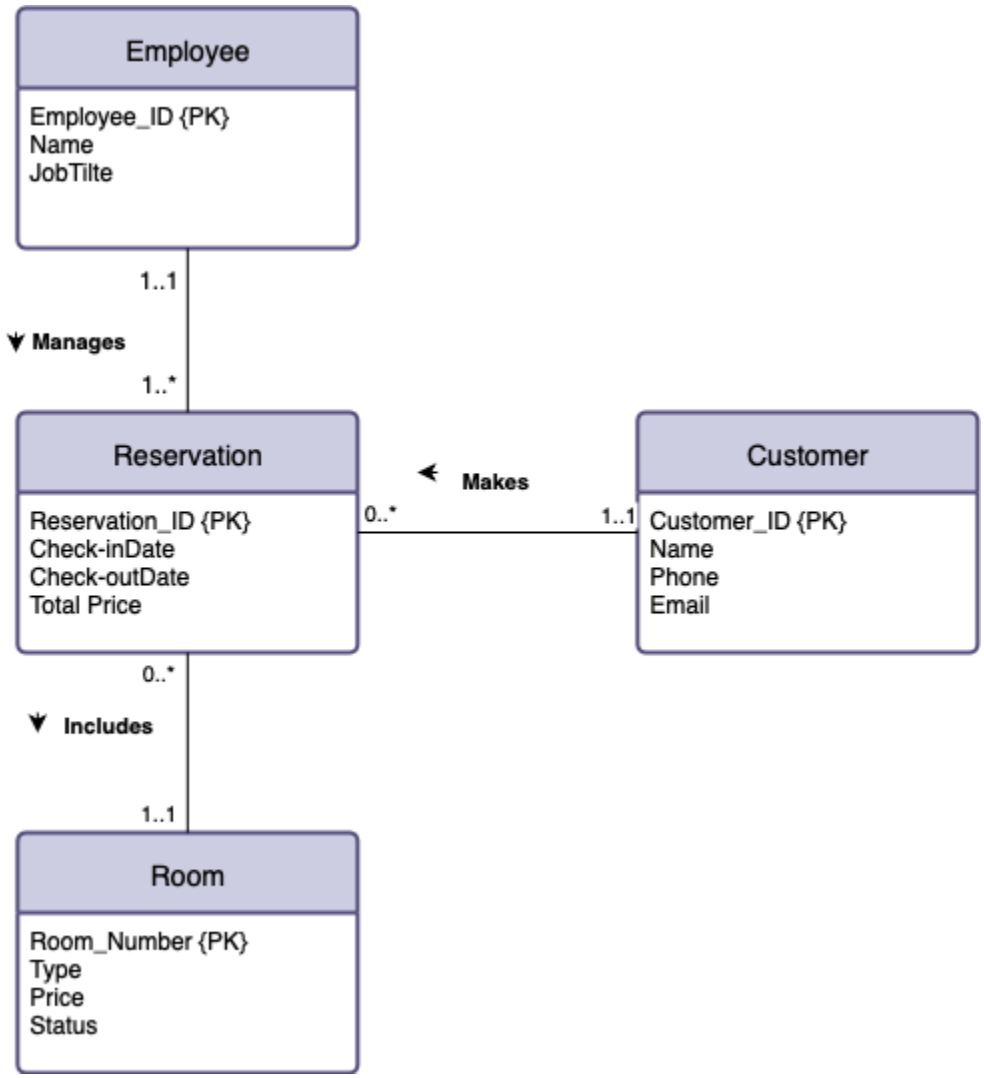
Customer:
The person who will book the room. The customer has a unique Customer_ID, name, phone, and email. Each customer can make (zero to many) reservations.

Room:
The room that can be booked by customers. The room has a unique Room_Number, type, price, and status (available or not). Each room can be linked to (zero to many) reservations.

Reservation :
The customer’s recorded booking information. It includes a unique Reservation_ID, check-in date, check-out date, and total price. Each reservation is connected to one customer and one room, also Reservation is linked to exactly one Employee.

Employee:
The hotel staff member who manages reservations. The employee has a unique Employee_ID, name, job title. One employee can handle many reservations.

Global enhanced entity relationship diagram (EER):



Relational Schema:

Map the Global EER to a relational schema and identify the Primary key and Foreign keys.

Customer (CustomerID, Name, Phone, Email)
Primary Key CustomerID

Room (RoomNumber, Type, Price, Status)
Primary Key RoomNumber

Employee (EmployeeID, Name, JobTitle)
Primary Key EmployeeID

Reservation(ReservationID, CheckIn, CheckOut, TotalPrice, CustomerID, RoomNumber, EmployeeID)

~~Primary Key Res~~ervationID
Foreign Key CustomerID references Customer(CustomerID)
Foreign Key RoomNumber references Room(RoomNumber)
Foreign Key EmployeeID references Employee(EmployeeID)

Data Dictionary showing description of all entities:

Entity Name	Description	Occurrence
Customer	Represents the person who makes a reservation. Includes unique Customer_ID, name, phone number, and email	Each Customer can make zero to many Reservations. Each Reservation is made by exactly one Customer

Room	Represents the rooms available for booking in the hotel. Each room has a unique Room_Number, type, price, and status	Each Room can be included in zero to many Reservations. Each Reservation includes exactly one Room
Reservation	Represents the booking details, including check_in and check_out dates, total price, and unique Reservation_ID. It links Customers, Employees, and Rooms.	Each Reservation is made by one Customer. Each Reservation is managed by one Employee.- Each Reservation includes one Room
Employee	Represents the hotel staff who manage reservations. Includes unique Employee_ID, name, job title.	Each Employee manages one or more Reservations. Each Reservation is managed by exactly one Employee

Data Dictionary showing description of all relationships:

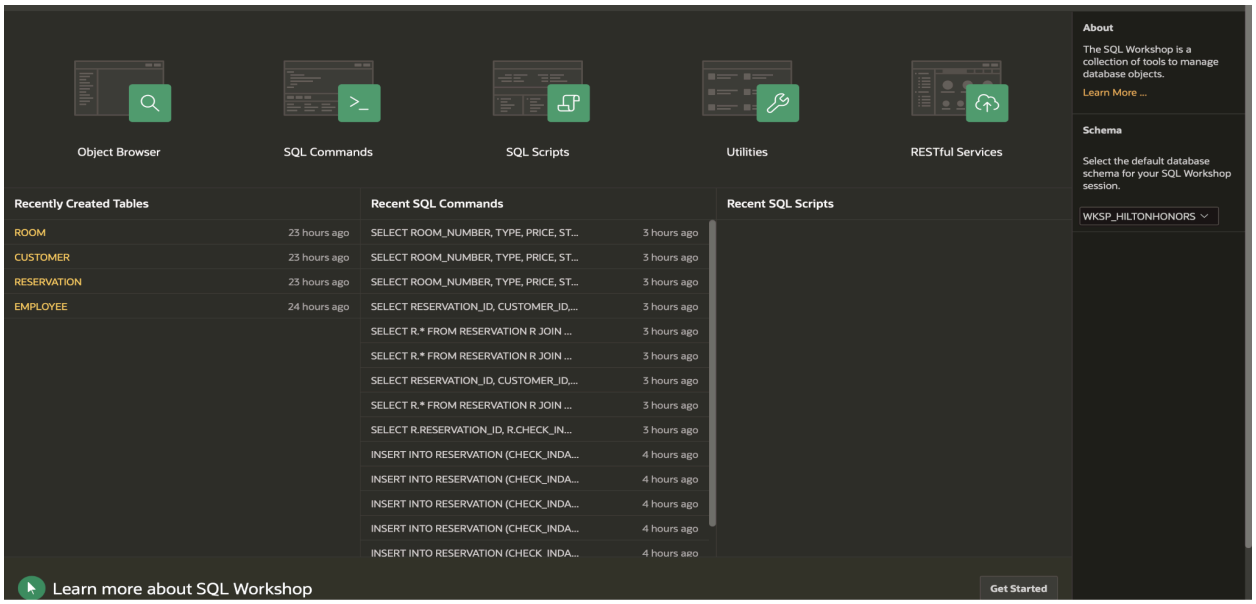
Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Customer	**1..1**	**Makes**	**Reservation**	**0..***
Room	**1..1**	**Included in**	**Reservation**	**0..***
Employee	**1..1**	**Manages**	**Reservation**	**1..***

Data Dictionary showing description of all attributes:

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
Employee	Employee_ID	Unique identifier for each employee	Integer	10	No	No	-	-	Yes
	Name	Full name of the employee	Varchar	50	No	No	-	-	No
	JobTitle	The employee's job title or position	Varchar	30	Yes	No	-	-	No
Reservation	Reservation_ID	Unique identifier for each reservation	Integer	10	No	No	-	-	Yes
	Check_inDate	The date when the reservation starts	Date	-	No	No	-	Valid date	No
	Check-out Date	The date when the reservation ends	Date	-	No	No	-	Valid date	No

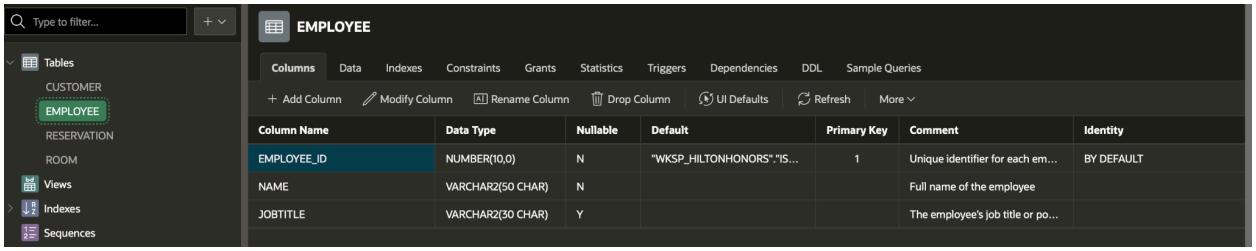
	TotalPrice	The total price for the reservation	Decimal	10,2	No	No	0	0<=	No
Customer	Customer_ID	Unique identifier for each customer	Integer	—	No	No	Auto-increment	> 0	Yes
	Name	Full name of the customer	Varchar	50	No	No	—	1–50 chars (letters/spaces)	No
	Phone	Customer phone number	Varchar	10	No	No	—	E.g., +9665XXXXXXXX (8–15 digits/+)	No
	Email	Customer email address	Varchar	100	No	No	—	valid email format (e.g., name@domain.com)	No
Room	Room_Number	Unique number for each room	Integer	—	No	No	—	> 0	Yes
	Type	Room category	Varchar	20	No	No	—	Must be one of: Single, Double, Suite (project scope)	No
	Price	Nightly price	Decimal	10,2	No	No	0.00	≥ 0.00	No
	Status	Availability state	Varchar	15	No	No	available	Allowed: available, occupied, maintenance	No

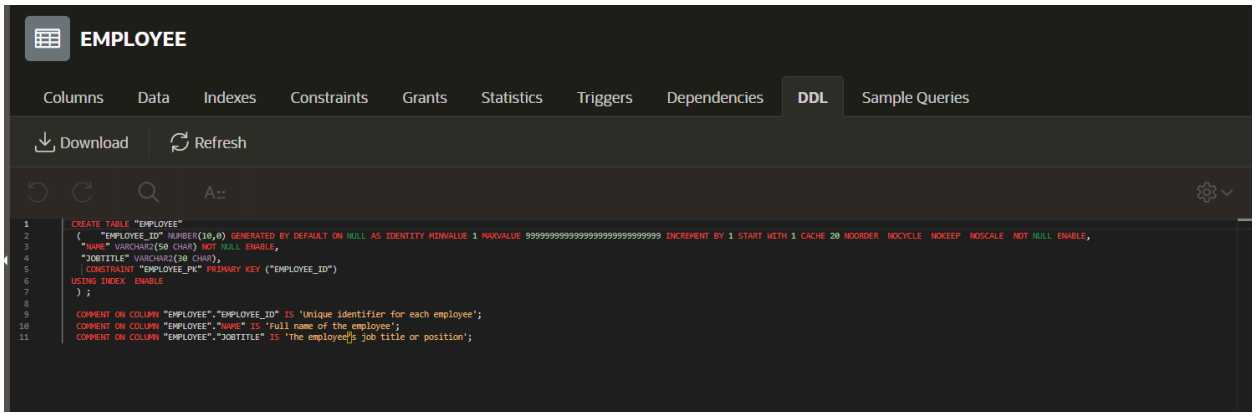
SQL Workshop



DB tables creation:

EMPLOYEE





CREATE TABLE "EMPLOYEE"

("EMPLOYEE_ID" NUMBER(10,0) GENERATED BY DEFAULT ON NULL AS IDENTITY MINVALUE 1 MAXVALUE 9999999999999999999999999999 INCREMENT BY 1 START WITH 1 CACHE 20 NOORDER NOCYCLE NOKEEP NOSCALE NOT NULL ENABLE,

"NAME" VARCHAR2(50 CHAR) NOT NULL ENABLE,

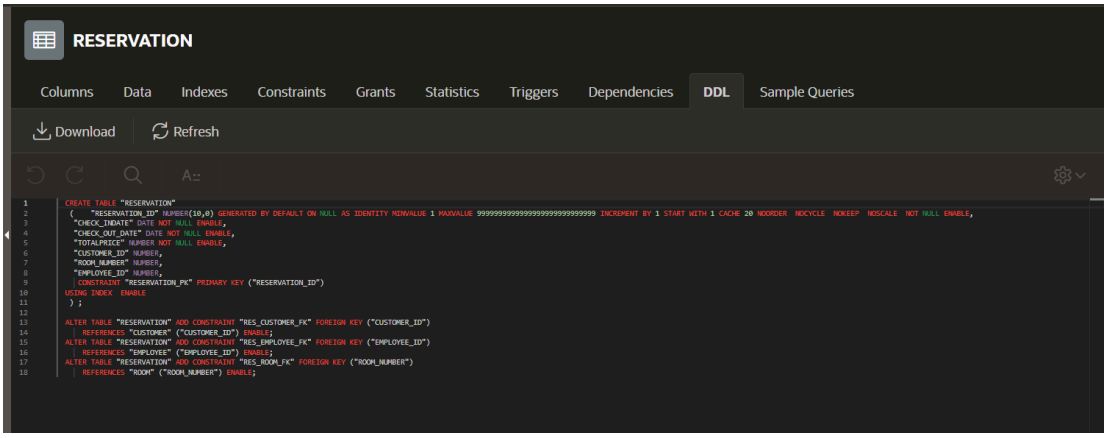
"JOBTITLE" VARCHAR2(30 CHAR),

CONSTRAINT "EMPLOYEE_PK" PRIMARY KEY ("EMPLOYEE_ID")

USING INDEX ENABLE

) ;

RESERVATION



CREATE TABLE "RESERVATION"

("RESERVATION_ID" NUMBER(10,0) GENERATED BY DEFAULT ON NULL AS IDENTITY MINVALUE 1 MAXVALUE 9999999999999999999999999999 INCREMENT BY 1 START WITH 1 CACHE 20 NOORDER NOCYCLE NOKEEP NOSCALE NOT NULL ENABLE,

"CHECK_INDATE" DATE NOT NULL ENABLE,

"CHECK_OUT_DATE" DATE NOT NULL ENABLE,

"TOTALPRICE" NUMBER NOT NULL ENABLE,

"CUSTOMER_ID" NUMBER,

"ROOM_NUMBER" NUMBER,

"EMPLOYEE_ID" NUMBER,

CONSTRAINT "RESERVATION_PK" PRIMARY KEY ("RESERVATION_ID")

USING INDEX ENABLE

) ;

ALTER TABLE "RESERVATION" ADD CONSTRAINT "RES_CUSTOMER_FK" FOREIGN KEY ("CUSTOMER_ID")

REFERENCES "CUSTOMER" ("CUSTOMER_ID") ENABLE;

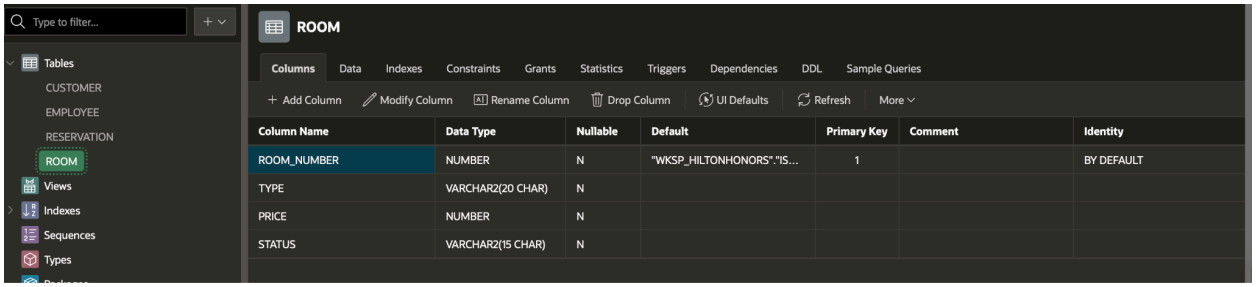
ALTER TABLE "RESERVATION" ADD CONSTRAINT "RES_EMPLOYEE_FK" FOREIGN KEY ("EMPLOYEE_ID")

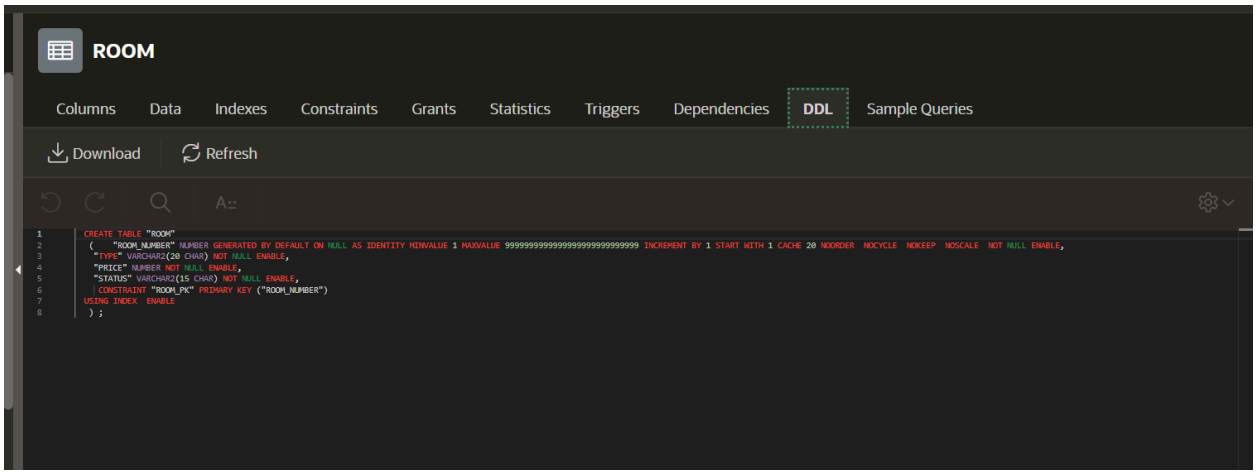
REFERENCES "EMPLOYEE" ("EMPLOYEE_ID") ENABLE;

ALTER TABLE "RESERVATION" ADD CONSTRAINT "RES_ROOM_FK" FOREIGN KEY ("ROOM_NUMBER")

REFERENCES "ROOM" ("ROOM_NUMBER") ENABLE;

ROOM





CREATE TABLE "ROOM"

("ROOM_NUMBER" NUMBER GENERATED BY DEFAULT ON NULL AS IDENTITY MINVALUE 1 MAXVALUE 9999999999999999999999999999 INCREMENT BY 1 START WITH 1 CACHE 20 NOORDER NOCYCLE NOKEEP NOSCALE NOT NULL ENABLE,

"TYPE" VARCHAR2(20 CHAR) NOT NULL ENABLE,

"PRICE" NUMBER NOT NULL ENABLE,

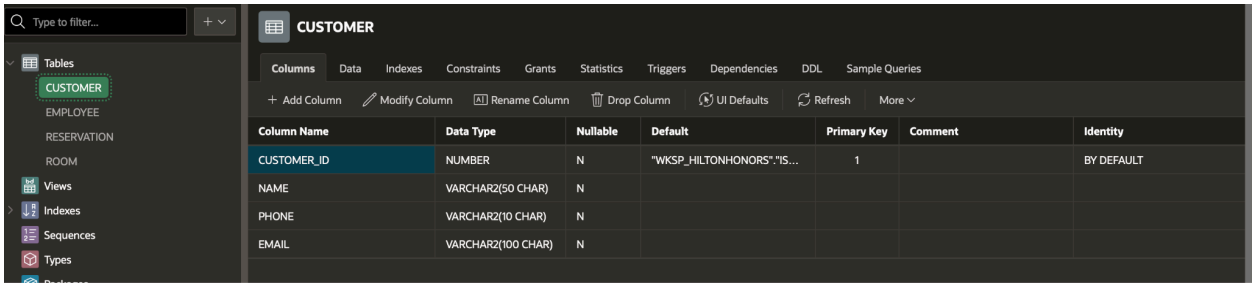
"STATUS" VARCHAR2(15 CHAR) NOT NULL ENABLE,

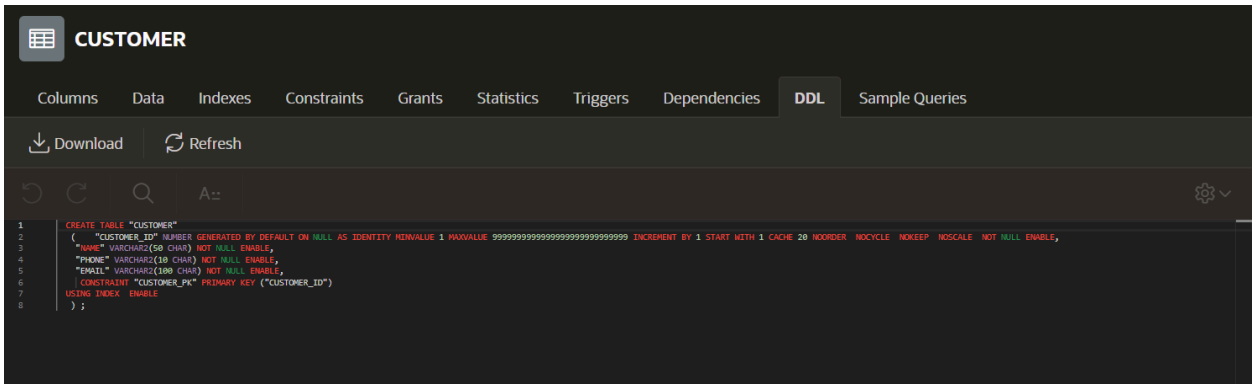
CONSTRAINT "ROOM_PK" PRIMARY KEY ("ROOM_NUMBER")

USING INDEX ENABLE

) ;

CUSTOMER





CREATE TABLE "CUSTOMER"

("CUSTOMER_ID" NUMBER GENERATED BY DEFAULT ON NULL AS IDENTITY MINVALUE 1 MAXVALUE 9999999999999999999999999999 INCREMENT BY 1 START WITH 1 CACHE 20 NOORDER NOCYCLE NOKEEP NOSCALE NOT NULL ENABLE,

"NAME" VARCHAR2(50 CHAR) NOT NULL ENABLE,

"PHONE" VARCHAR2(10 CHAR) NOT NULL ENABLE,

"EMAIL" VARCHAR2(100 CHAR) NOT NULL ENABLE,

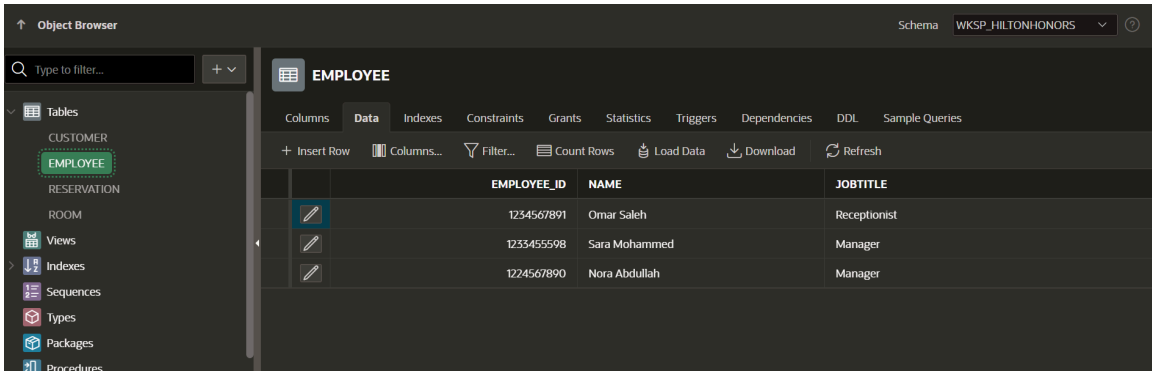
CONSTRAINT "CUSTOMER_PK" PRIMARY KEY ("CUSTOMER_ID")

USING INDEX ENABLE

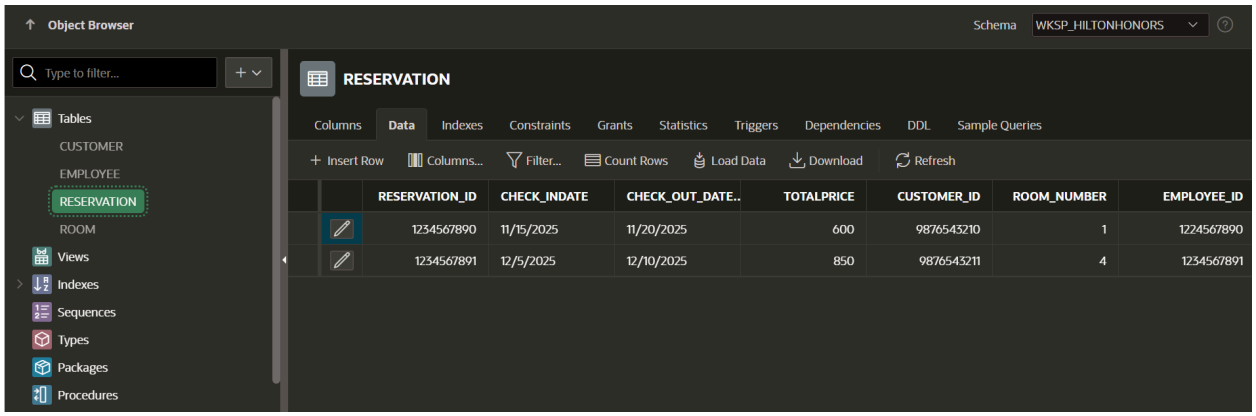
) ;

Data insertion:

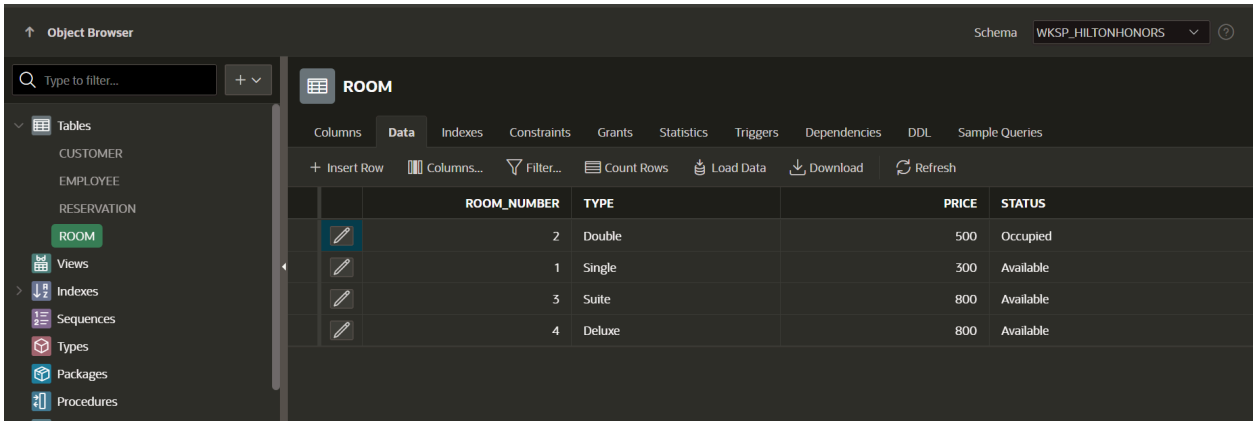
EMPLOYEE



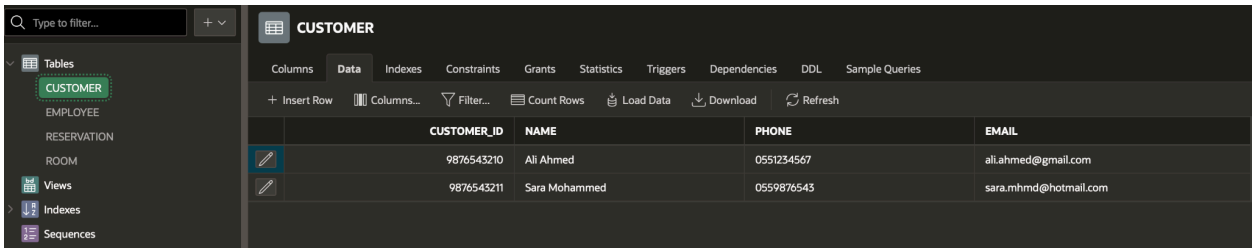
RESERVATION



ROOM

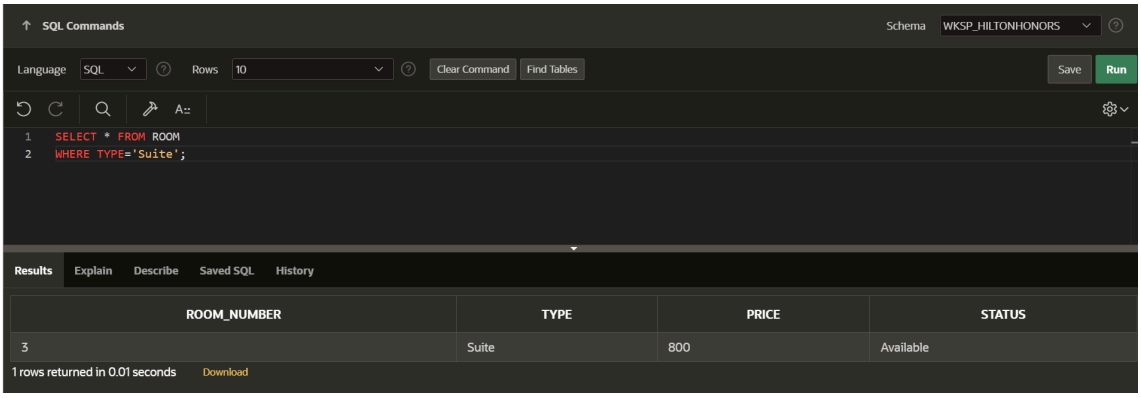


CUSTOME

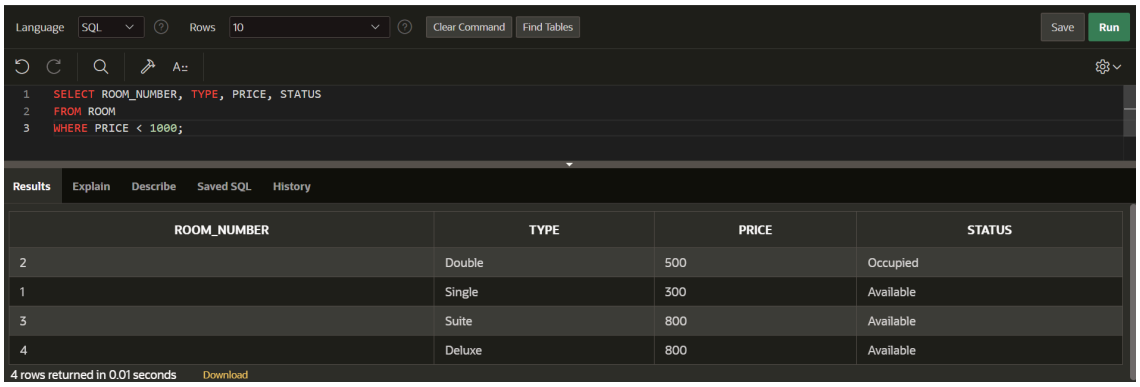


Data Queries:

1- List all rooms of type ‘Suite



2- List rooms with prices less than 1000



3- List my reservations for a specific room type and price range

```
SELECT R.*
FROM RESERVATION R
JOIN ROOM RM ON R.ROOM_NUMBER = RM.ROOM_NUMBER
WHERE RM.TYPE='Deluxe'
  AND R.TOTALPRICE BETWEEN 500 AND 1000;
```

RESERVATION_ID	CHECK_INDATE	CHECK_OUT_DATE	TOTALPRICE	CUSTOMER_ID	ROOM_NUMBER	EMPLOYEE_ID
1234567891	12/5/2025	12/10/2025	850	9876543211	4	1234567891

1 rows returned in 0.01 seconds

4- List all rooms ordered by price from lowest to highest.

```
SELECT ROOM_NUMBER, TYPE, PRICE, STATUS
FROM ROOM
ORDER BY PRICE ASC;
```

ROOM_NUMBER	TYPE	PRICE	STATUS
1	Single	300	Available
2	Double	500	Occupied
4	Deluxe	800	Available
3	Suite	800	Available

4 rows returned in 0.01 seconds

5- Update the phone number of a guest.

```
UPDATE CUSTOMER
SET PHONE = '0559999999'
WHERE NAME = 'Ali Ahmed';
```

1 row(s) updated.

0.03 seconds

Schema WKSP_HILTONHONORS

CUSTOMER

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

CUSTOMER_ID	NAME	PHONE	EMAIL
9876543210	Ali Ahmed	0559999999	ali.ahmed@gmail.com
9876543211	Sara Mohammed	0559876543	sara.mhmd@hotmail.com

6- The guest enters his name,phone, and emails.

Language SQL | Rows 10 | Clear Command | Find Tables | Save | Run

```
INSERT INTO CUSTOMER (NAME, PHONE, EMAIL)
VALUES ('Raneem Mohammed', '0570701546', 'Raneem@gmail.com');
```

Results | Explain | Describe | Saved SQL | History

1 row(s) inserted.

0.02 seconds

Schema WKSP_HILTONHONORS

CUSTOMER

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

CUSTOMER_ID	NAME	PHONE	EMAIL
9876543210	Ali Ahmed	0559999999	ali.ahmed@gmail.com
21	Raneem Mohammed	0570701546	Raneem@gmail.com
9876543211	Sara Mohammed	0559876543	sara.mhmd@hotmail.com

7-List rooms with prices greater than average

Language SQL Rows 10 Clear Command Find Tables Save Run

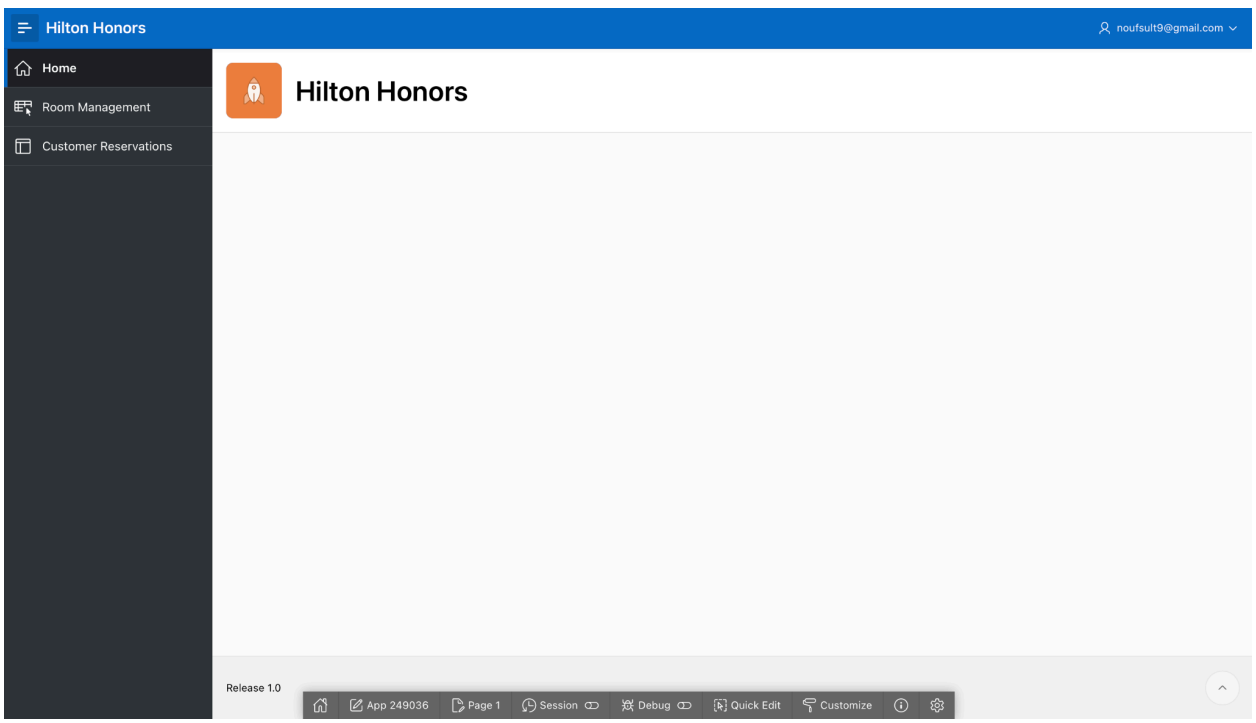
```
SELECT ROOM_NUMBER, TYPE, PRICE, STATUS
FROM ROOM
WHERE PRICE > (SELECT AVG(PRICE) FROM ROOM);
```

Results Explain Describe Saved SQL History

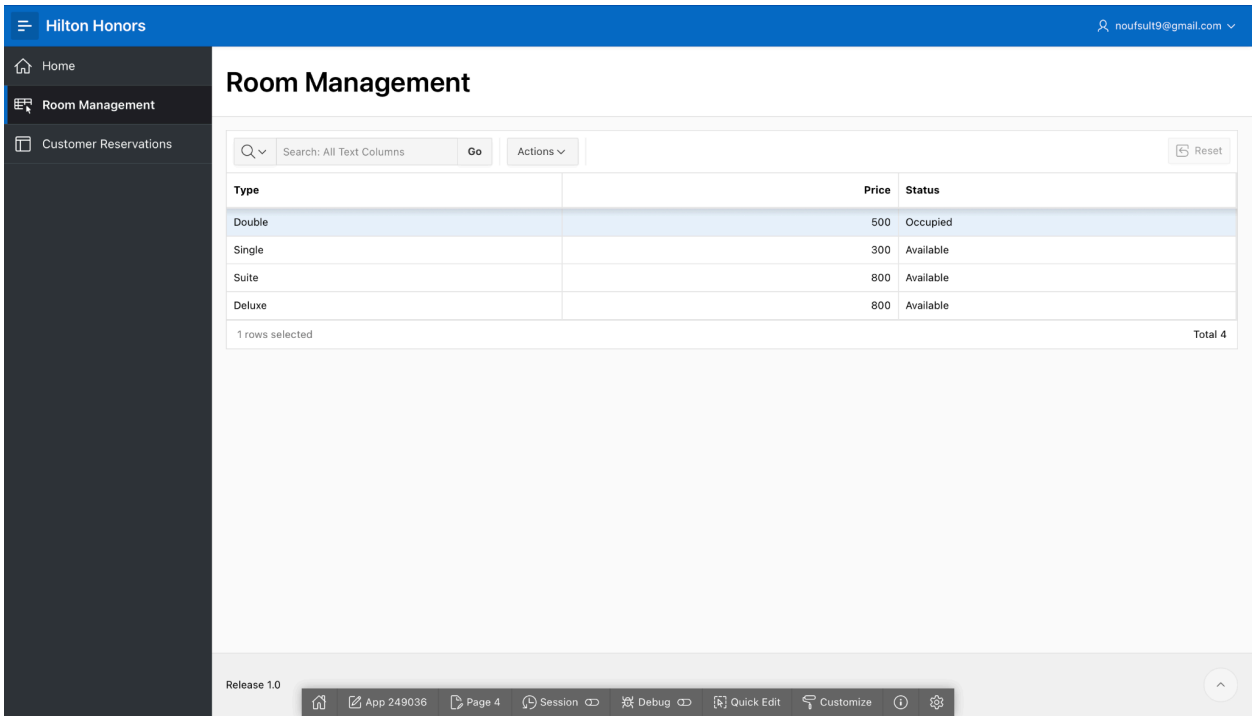
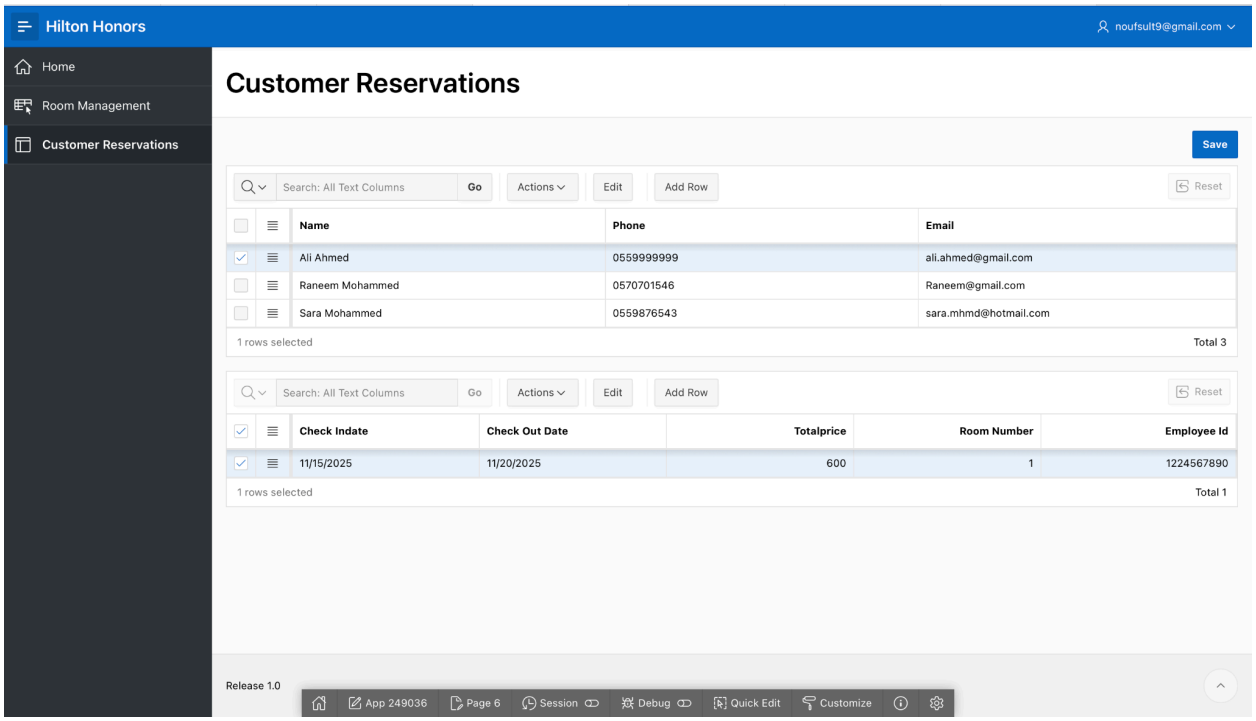
ROOM_NUMBER	TYPE	PRICE	STATUS
3	Suite	800	Available
4	Deluxe	800	Available

2 rows returned in 0.01 seconds Download

App Builder:



Interactive Grid:



Master Detail: